

Adentro de Widevine L1 en Windows: El despacho de tres capas que nadie mapeó, y las ranuras por-cuadro que nunca disparan

Jeremy Erazo (trexnegr0)
Investigador independiente de seguridad
mendozayt13@gmail.com

Junio de 2026

Resumen ejecutivo. El CDM Widevine L1 que viene en todo navegador de la familia Chromium en Windows ha sido, hasta ahora, una caja negra para cualquiera fuera de Google. Este paper la abre. Tres vtables concéntricas, diecinueve ranuras públicas de CDM_11, verificadas byte-a-byte en dos builds de navegadores distintos. Un mapa empírico runtime-confirmado de qué ranuras se ejercen realmente durante streaming hardware-decoded real. Una distribución de ofuscación estilo MBA a lo largo del espacio de ranuras que refleja la profundidad del parsing de datos atacante-controlables. Tablas de alcanzabilidad tanto para Netflix como para una fuente pública Widevine no-Netflix. Y, de fondo, una metodología que captura todo esto sin jamás leer un byte de contenido, una clave de contenido, ni un IV desde la memoria del CDM. Los defensores se llevan una tabla de superficie de ataque por ranura más firmas de detección; los atacantes se llevan la desambiguación de cuáles ranuras tienen que vivirles. La metodología completa se libera con fuente junto al paper; no se libera artefacto de extracción ni exploit weaponizado.

Resumen

El Módulo de Descifrado de Contenido (CDM) Widevine viene prácticamente en todo navegador de escritorio mayor y soporta el camino de reproducción de Encrypted Media Extensions (EME) para todo servicio comercial de streaming. Los internals del build L1 de Windows están documentados sólo fragmentariamente: los archivos de cabecera describen el ABI de C++ que expone al navegador anfitrión, pero el binario en sí es código cerrado, parcialmente ofuscado, y se comporta diferente dependiendo de si el anfitrión delega la decodificación a pipelines protegidos por hardware. Este paper presenta una ingeniería inversa arquitectural estilo Buchanan de la implementación CDM_11 tal como viene en Chrome 149 y Edge 150 sobre Windows 11. Identificamos un *modelo de despacho de tres capas* (*thunks* de la vtable externa, una vtable intermedia `m_impl` de *wrappers* limpios y *double-trampolines*, y cuerpos de *handlers* por ranura), localizamos cada capa byte-a-byte en ambos binarios, y caracterizamos empíricamente cuáles ranuras se invocan durante reproducción en steady-state vs. cuáles permanecen dormidas. Confirmamos en runtime que durante reproducción hardware-decoded de Netflix las ranuras por-cuadro del CDM_11 (`Decrypt`, `DecryptAndDecodeFrame`, `DecryptAndDecodeSamples`) no disparan mientras una tabla de despacho singleton interna genera ~2,300 llamadas por segundo de trabajo cripto/scheduler invisible desde el límite EME. Documentamos la distribución de ofuscación estilo mixed-boolean-arithmetic en el espacio de ranuras y mostramos que se concentra donde el CDM empieza a parsear datos estructurados atacante-controlables, mientras está ausente en ranuras que pasan *blobs* opacos. Después verificamos la alcanzabilidad de las ranuras de gestión-de-sesión desde dos aplicaciones

EME distintas (Netflix y una fuente pública de prueba Widevine) y mostramos comportamiento de despacho idéntico. El paper sitúa este trabajo en dos décadas de investigación previa de DRM, recorre los precedentes legales y metodológicos, y documenta tanto los caminos exitosos como los falsos arranques que caracterizaron el proceso de ingeniería inversa. Ningún clave de contenido, buffer de cuadro, ni byte de ciphertext se captura en ningún momento del estudio.

1. Introducción

Modern desktop video DRM is the bridge between commercial streaming services and the browser’s media pipeline. When a user plays a video in Netflix, Amazon Prime, Disney+ or any other Widevine-protected source, the browser invokes Encrypted Media Extensions (EME) to delegate the cryptographic operations to a Content Decryption Module (CDM). In Chromium-based browsers on Windows, that CDM is the proprietary `widevinecdm.dll` library shipped by Google [1]. The browser loads the library inside a Mojo utility process (`-utility-sub-type=media.mojom.`

`CdmServiceBroker`), connects to it over IPC, and asks the library to construct a CDM instance implementing a stable C++ interface defined in `content_decryption_module.h` [2]. From the host’s point of view the interface is small (nineteen virtual methods) and well documented: `Initialize`, `GetStatusForPolicy`, `CreateSessionAndGenerateRequest`, `UpdateSession`, and twelve other methods that cover license exchange, decryption, decode, and lifecycle.

From the *library’s* point of view this clean nineteen-method surface is the visible peak of a much larger iceberg. The Windows L1 build of Widevine implements a multi-million-instruction obfuscated runtime under tens of megabytes of closed source code, integrates with Windows Protected Media Path (PMP) when hardware decode is available, and is bound by a strict licensing regime that prohibits study of its decryption internals. Public knowledge of the implementation is fragmentary: the Chromium tree contains the *host-side* of the EME plumbing, several academic and hobbyist projects have studied the *Android L3* variant [19, 21], but the desktop Windows L1 variant has received comparatively little architectural attention in the open literature.

The motivation for this study is methodological as much as it is technical. Several recent legal episodes have shown that *publishing what one learns* about DRM internals is more legally exposed than *learning it*, and that the reproducibility of a result matters as much as the result itself [15, 16, 18]. Following the Buchanan model of architectural reverse engineering [10, 11] — probe everything, document the technique, publish no extraction artefact — we set out to build a fully reproducible, fully open methodology for studying the Widevine CDM and to apply it to the questions that are paper-able without crossing the digital-rights-management circumvention line.

Lo nuevo de este trabajo

1. **The first public architectural map of L1 `widevinecdm.dll` on Windows.** Three concentric vtables, every entry classified, every dispatch offset byte-verified. Cross-validated in Chrome 149 and Edge 150. Drop-in reproducible against any future build via the released parameter-free discoverer.
2. **Hardware-decode dormancy, runtime-confirmed.** Four outer-vtable slots patched in a single experiment. Across 90+ seconds of continuous 4K HDR Netflix playback, `g_calls = 0` on slot 14 (`DecryptAndDecodeFrame`), slot 15 (`DecryptAndDecodeSamples`), and slot 9 (`Decrypt`) simultaneously. Per-frame work on L1 does not traverse the public CDM interface; the EME-facing attack surface for per-frame interception is, for practical purposes, empty.
3. **A per-slot attack-surface table.** Which slots ingest attacker-controllable structured input (PSSH, license response, policy), which slots pass opaque blobs, which slots are dormant. Cross-source consistent: Netflix and Bitmovin Widevine demo exercise the same slots in the same order. Defenders

use the table directly to scope detection; attackers use it directly to prioritise the layer-3 handlers that actually matter.

4. **MBA-style obfuscation, mapped depth-relative.** Each of the nineteen CDM_11 slots characterised at every layer of the dispatch. Obfuscation lives at varying depths, consistently appearing where structured attacker data begins to be parsed and consistently absent on passthrough slots. Concrete examples with byte-level signatures for each.
5. **A methodology that travels.** The *shape-only observation* pattern is generalisable to any closed-source binary with a stable interface. The same hook design applies, unchanged in shape, to PlayReady’s PlayReadyHost, FairPlay’s AppleSpringboard, or any other host-CDM contract where the question is *which entries fire when* and the answer must not capture content. We release the Buchanan-style framework as a methodology contribution in its own right.
6. **Defender’s playbook and attacker’s reading.** Two explicit sections that translate the findings into concrete action. Detection signatures, hook canaries, behavioural profiles, and per-slot prioritisation for both sides.
7. **Reproducible and falsifiable.** Every empirical claim anchors to a recorded experimental run. Every architectural claim is rederivable from the public binary by the released discoverer. The falsification sprint that preceded publication is documented in full.

Hoja de ruta del paper

The paper proceeds as follows. Section 2 situates this work in two decades of prior reverse engineering of commercial DRM and the legal precedents that have shaped it. Section 3 introduces the Widevine licensing levels, the EME dispatch path, the Mojo IPC architecture used by Chromium for CDM hosting, the Windows PMP boundary, the Common Encryption format (CENC) used by all Widevine content, and the PSSH box structure that carries attacker-controllable initialisation data. Section 4 states the threat model and ethical scope. Section 5 walks through the static reverse engineering of CDM_11 in detail, with concrete disassembly listings. Section 6 describes the runtime instrumentation, including the design subtleties of CIG bypass, CFG bitmap registration, and the per-slot back-pointer chain. Section 7 presents the empirical findings. Section 8 treats the obfuscation distribution as its own architectural defence. Section 9 establishes cross-binary and cross-source stability. Section 10 reports on the falsification tests, including the candidate findings that were refuted by deeper analysis. Section ?? discusses what the model implies for both attackers and defenders of desktop DRM. Section 14 states what was not done and why. Section 15 concludes.

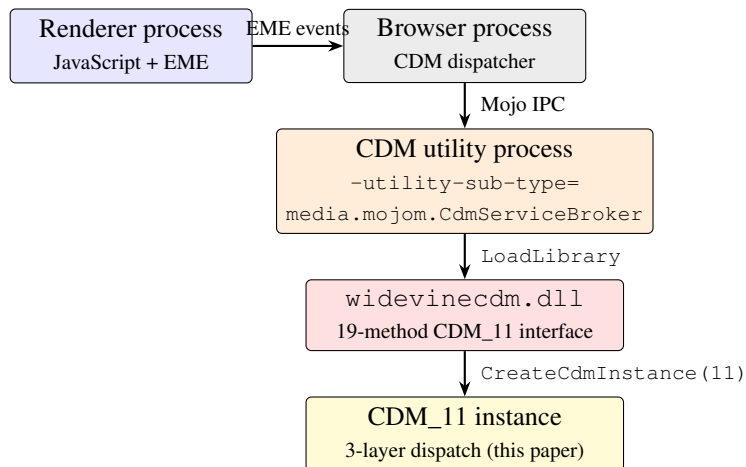


Figura 1: End-to-end Widevine EME dispatch on Chromium-family browsers on Windows. The host browser delegates Encrypted Media Extensions operations through Mojo IPC to a utility process whose sole purpose is to load `widevinecdm.dll` and expose its CDM_11 interface. This paper studies the dispatch model that sits between the `CreateCdmInstance`-returned interface and the per-slot handlers inside the library.

2. Antecedentes y trabajos relacionados

DRM reverse engineering occupies an unusual corner of security research: it spans applied cryptography, binary reverse engineering, copyright law, vendor relationships, and (sometimes) criminal prosecution. The methodology we follow in this paper did not emerge in a vacuum; it is shaped by two decades of prior work, some of which produced excellent technical results and some of which produced excellent technical results *and* adverse legal outcomes for the researcher. This section surveys the prior art and explains which precedents informed the methodological choices made here.

2.1. El modelo Buchanan

The methodological framework we follow is most clearly articulated in the recent work of David Buchanan [10, 11, 12]. Buchanan’s research notably includes the reverse engineering of Apple’s iMessage NaCl-based key wrap, the Apple T2 security chip’s secure boot chain, and an arbitrary-image bug in the iOS image decoder chain (CVE-2023-4863).

The defining property of Buchanan’s published work, for our purposes, is the strict separation of *architectural description* from *exploitable artefact*. His public output documents how a system is structured and where its trust boundaries sit; the controlled proofs of concept that accompany the writeups demonstrate the bug class without producing a tool that straightforwardly enables abuse against unrelated victims. He has publicly described the resulting publication discipline as: probe everything, document the technique, publish the architecture, do not publish the artefact that would let arbitrary readers materialise the impact.

For DRM research this distinction matters: 17 U.S.C. §1201 (the DMCA’s anti-circumvention provision) and its international equivalents prohibit the publication of tools or technologies whose *primary* use is the circumvention of a technological protection measure. An architectural description of a CDM is not such a tool. A frame ripper that consumes the same architectural description and produces decrypted Netflix MP4s plainly is. The gap is the line we are careful not to cross.

2.2. Hacking the Xbox y la tradición bunnie

Andrew “bunnie” Huang’s work on console reverse engineering — the original Xbox [13], the NeTV project’s HDCP-handling video pipeline [14], and the Novena laptop — demonstrated a sustainable model for DRM-adjacent research that was simultaneously useful, publishable, and not the subject of vendor litigation. The NeTV project in particular is significant: HDCP is a copyright-protection measure under U.S. law, and the project required EFF-coordinated correspondence with the Library of Congress to confirm that an architectural demonstration that does not capture, redistribute, or facilitate the redistribution of protected content remains lawful under the DMCA’s exemption framework.

Huang’s recurring observation is that the legal exposure attaches to the *exhibition* of a circumvention capability and the *distribution* of tools that perform it, not to the act of characterising a system. The CDM_11 dispatch model described in this paper would be obviously useful to an attacker who already possessed an independent capability to receive encrypted Widevine content and obtain valid licenses; it would be of essentially no use to a casual attacker who did not. That distribution of utility is, in our view, on the right side of the bunnie line.

2.3. Sklyarov, DVD-Jon, y la frontera legal

The cases of **Dimitry Sklyarov** and **Jon Lech Johansen** bracket the legal frontier of DRM research.

Sklyarov, a Russian developer for ElcomSoft, was arrested in 2001 at DEF CON for his work on the Advanced eBook Processor, which could convert Adobe’s protected eBooks into PDF [15]. The prosecution proceeded against ElcomSoft (Sklyarov was eventually released after agreeing to testify). The arrest, the trial, and the eventual acquittal of ElcomSoft on the criminal charges established two relevant precedents: (i) prosecutors will treat a tool whose output is the cleartext form of a protected work as “primarily used to circumvent,” even if the underlying research is academically defensible; (ii) merely *describing* the circumvention technique, without distributing a tool that performs it, was not the basis for the prosecution.

Jon Lech Johansen, “DVD-Jon,” was prosecuted by Norwegian authorities (acquitted twice, eventually finally) for his role in the development of DeCSS, the tool that broke CSS scrambling on DVDs [16]. The Norwegian courts held that Johansen’s work on his *own* DVD player was lawful, even though the resulting tool was redistributed widely on the internet. The U.S. case of *Universal v. Reimerdes* held that the *posting* of DeCSS on a website was actionable circumvention of the DMCA, regardless of the lawfulness of the underlying research [17].

The composite lesson from these cases informs our publication discipline: an architectural description is not a tool; a tool that materialises decrypted content is a tool. We publish the architectural description and the controlled validation framework that proves the description correct; we do not publish a tool.

2.4. Trabajo académico sobre Widevine

The Widevine ecosystem has received attention from several academic and corporate-research groups, primarily focused on the Android L3 variant.

Quarkslab published a series of analyses of Widevine on Android beginning in 2018 [19, 20]. Their work characterised the OEMCrypto-protected components of the Widevine SDK and demonstrated that the L3 keybox-based protection model relied substantially on software obfuscation that could be peeled back with whitebox-crypto attack techniques. The L3 decryptor projects [21, 22] that followed — mostly published as research-grade open-source — demonstrated end-to-end Widevine L3 content extraction from the protected library. Google’s response was to deprecate L3 web-browser support and accelerate the migration to L1.

Check Point Research has periodically analysed Widevine on Android handsets, identifying specific keybox extraction vulnerabilities and bug classes in the OEMCrypto TrustZone [23]. Their reports were coordinated with Google and the vendor before publication.

Romain Thomas and collaborators at Quarkslab have published extensively on practical reverse engineering tools that we use here, including the LIEF library for cross-platform binary parsing [24] and several blog posts on whitebox crypto in commercial DRM. Their tool work is directly upstream of the methodology used in this paper for static analysis of `widevinecdm.dll`.

There is, to our knowledge, no published academic study of the Windows L1 CDM dispatch internals as such. The work nearest in spirit is a handful of blog posts and forum threads that document specific issues (e.g., the `libwidevinecdm.so` symbol layout on Linux, the CDM ABI changes between major versions) without proposing a generalisable architectural model. This is the gap the present paper attempts to fill for the Windows L1 build.

2.5. Criptografía whitebox y ofuscación MBA

The obfuscation patterns observed in CDM_11 layer-3 handlers — high-entropy magic constants written to the stack at function entry, multiplicative-hash state-advance instructions, branchless `cmovae` cascades, `0xaaaaaaaa` poison fills — match the textbook signature of *mixed-boolean-arithmetic* (MBA) obfuscation as treated in the academic literature [25, 26]. MBA obfuscation transforms an arithmetic operation into an equivalent expression that mixes bitwise and arithmetic operations in a way that defeats algebraic simplification on the obfuscated form. It is widely used in commercial software protection products (VMProtect, Themida, Arxan) and in custom obfuscators built by vendors for specific defensive use cases.

A distinct but related technique, *whitebox cryptography*, embeds cryptographic key material into table-based representations that resist direct extraction even when the attacker has full read access to the obfuscated code [27, 28]. Commercial DRM products use whitebox-AES variants extensively; the encrypted keys in a Widevine L3 license are unwrapped by exactly such a routine before being applied to the per-frame decryption.

The pattern we report in Section 8 — obfuscation concentrated at the depth where structured attacker data begins to be parsed — is consistent with the design prescription of the whitebox literature: protect the cryptographic processor, not the dispatch glue around it. Whether the protection *succeeds* against a determined attacker is a separate question we do not address.

2.6. Antecedentes de herramientas

Three concrete pieces of tooling that this work relies on deserve acknowledgement:

Capstone [29], the multi-architecture disassembly framework used by every script in this release that walks `widevinecdm.dll`'s `.text` section.

pefile [30], the pure-Python PE parser used to enumerate exports, walk the section table, and read RVAs from `.rdata`.

mingw-w64, the Windows cross-toolchain used to build the shim and the helpers from a Linux host. Notably, mingw-w64's gcc backend does not support Microsoft's structured exception handling `__try / __except`, which forced us to use `VirtualQuery`-based safety checks for the shim's register-target read paths instead of SEH-protected `memcpy`. This is the kind of small implementation detail that doesn't appear in glossy DRM-research literature but determines whether the methodology runs at all.

3. Marco técnico

This section reviews the major technical components of the Windows L1 Widevine playback pipeline in greater depth than the introduction permitted.

3.1. Niveles de licencia de Widevine

Widevine defines three nested security levels.

L3 is a pure-software CDM. Cryptographic state is held in process memory inside an obfuscated runtime. Content keys are unwrapped by a whitebox-style cryptographic core and applied to ciphertext segments in user-space buffers. Decrypted output is returned to the host through standard memory. Per the published Widevine licensing policy, L3-only clients top out at 540p; content owners' deals frequently restrict L3 to lower bitrates than L2/L1 clients receive.

L2 is a hybrid: a software CDM holds at least its sensitive operations behind a hardware-backed Trusted Execution Environment (TEE) on the host. L2 sees limited deployment on desktop.

L1, the variant studied in this paper, requires the host to provide a hardware-backed protected media pipeline. On Windows the relevant pipeline is Protected Media Path (PMP), itself a composite of:

- `peauth.sys`, the Microsoft-signed kernel-mode component that enforces the OPM contract between the CDM and the GPU driver,
- the Media Foundation `IDirectXVideoDecoder` chain running in a kernel-protected mode,
- the DXGI protected-surface allocator that exposes the decoded frame only to the GPU presentation pipeline and not to user-space readback.

Under L1 hardware decode, the decrypted frame never appears in user-mode browser process memory. This is the design property that the paper's first empirical finding (Section 7.1) exploits to demonstrate the CDM dormancy phenomenon.

3.2. Encrypted Media Extensions and the dispatch path

The W3C Encrypted Media Extensions specification [7] defines the JavaScript API by which a web application requests DRM-mediated playback of an encrypted media stream. From the host browser's point of view, the relevant operations are:

1. `navigator.requestMediaKeySystemAccess("com.widevine.alpha")`: confirm that the client has a CDM for the named key system.
2. Create a `MediaKeys` object and bind it to an `HTMLMediaElement`.
3. Wait for the `encrypted` event, which provides the Common Encryption (CENC) initialisation data extracted from the media stream.
4. Create a `MediaKeySession` and call `generateRequest(initDataType, initData)`, which causes the CDM to construct a license request.
5. Forward the resulting message to the licensing server, receive the response, and feed it back via `updateSession(licenseResponse)`.
6. Continue playback as the CDM unwraps content keys and applies them to the encrypted media data.

Chromium’s implementation of this API routes each step through the browser process, the renderer, and the CDM utility process via Mojo IPC interfaces defined in `chromium/components/cdm` [3]. The relevant Mojo interface for our purposes is `media.mojom.ContentDecryptionModule`, whose request and response messages are essentially a serialised projection of the nineteen-method CDM_11 C++ interface.

3.3. The CDM_11 C++ interface

The interface CDM_11 implements is defined in the Chromium tree at `media/cdm/api/content_decryption_module.h`. The class is a pure-virtual base `ContentDecryptionModule_11` with the following nineteen entries, declared in this order:

```

1 virtual void Initialize(...) = 0;
2 virtual void GetStatusForPolicy(...) = 0;
3 virtual void SetServerCertificate(...) = 0;
4 virtual void CreateSessionAndGenerateRequest(...) = 0;
5 virtual void LoadSession(...) = 0;
6 virtual void UpdateSession(...) = 0;
7 virtual void CloseSession(...) = 0;
8 virtual void RemoveSession(...) = 0;
9 virtual void TimerExpired(...) = 0;
10 virtual Status Decrypt(...) = 0;
11 virtual Status InitializeAudioDecoder(...) = 0;
12 virtual Status InitializeVideoDecoder(...) = 0;
13 virtual void DeinitializeDecoder(...) = 0;
14 virtual void ResetDecoder(...) = 0;
15 virtual Status DecryptAndDecodeFrame(...) = 0;
16 virtual Status DecryptAndDecodeSamples(...) = 0;
17 virtual void OnPlatformChallengeResponse(...) = 0;
18 virtual void OnQueryOutputProtectionStatus(...) = 0;
19 virtual void OnStorageId(...) = 0;

```

Each virtual call translates, under the Microsoft x64 calling convention, into a single indirect call through a function pointer loaded from the object’s vtable. The vtable layout for a class with nineteen virtual functions and no base-class virtuals is a nineteen-entry contiguous array of code pointers; the per-call overhead from the host’s perspective is one indirect load plus the CFG dispatcher cost.

We adopt the convention of identifying CDM_11 methods by their slot index throughout this paper. Decrypt is slot 9, DecryptAndDecodeFrame is slot 14, and so on.

3.4. Common Encryption (CENC) and PSSH

The Widevine protocol carries its initialisation data inside a Protection-Specific System Header (PSSH) box, as standardised by ISO/IEC 23001-7 (Common Encryption for ISO Base Media File Format) [9, 8]. A PSSH box is structured:

```

1 PSSHBox {
2     u32          size;                // big-endian
3     char[4]      type;                // "pssh"
4     u8           version;
5     u24          flags;
6     UUID(16B)    SystemID;            // Widevine SystemID
7                                     // edef8ba9-79d6-4ace-a3c8-27cd51d21ed
8     if version > 0:
9         u32      KID_count;
10        UUID[KID_count] KIDs;

```

```

11     u32         data_size;
12     u8[data_size] data;
13 }

```

The `data` field, for Widevine, contains a Protocol Buffers serialisation of a `WidevineCencHeader` message with a known schema. The `generateRequest` call passes the entire PSSH box to the CDM, which parses out the `SystemID`, the KIDs, and the protobuf-encoded header to construct a license request.

This is the attacker-controllable structured input that the `CreateSessionAndGenerateRequest` slot (slot 3) ingests: a malicious DASH manifest or a malicious EME application can place arbitrary bytes inside the PSSH and pass them to the CDM with full control over both the box header fields and the protobuf payload. Parser vulnerabilities in the CDM's PSSH path are attacker-reachable from any compromised CDN that serves the DASH manifest, from any compromised JavaScript that drives the EME application, or from any MITM attacker who can rewrite the DASH manifest in transit.

The two standard CENC encryption schemes that the CDM must support are **CENC-CTR** (AES-128 in Counter mode, with subsample descriptors that indicate which byte ranges of the encrypted track are protected) and **CBCS** (AES-128 in CBC mode with *pattern* encryption: a 1-in-10 or 1-in-9 pattern of encrypted-skipped blocks designed to be tolerable to AVC/HEVC NALU boundaries). The `CDM_11 InputBuffer_2` structure carries both schemes and their subsample / pattern parameters per call.

3.5. Mojo IPC and the CDM utility process

Chromium isolates the CDM in a dedicated utility process to confine the impact of CDM compromise. The utility process is spawned with the command line:

```

1 chrome.exe --type=utility
2             --utility-sub-type=media.mojom.CdmServiceBroker
3             --lang=en-US --service-sandbox-type=cdm
4             --no-sandbox (when CIG is disabled for research)
5             ...

```

Inside the utility, a Mojo interface implementation accepts the host's CDM requests and forwards them to a C++ class that owns a single `ContentDecryptionModule_11` pointer. That pointer points at the object returned by `widevinecdm.dll!CreateCdmInstance(11, ...)` — which is the object whose vtable we map in Section 5.

3.6. Control Flow Guard and Code Integrity Guard

Two Windows process-level mitigations are relevant to the runtime instrumentation of the CDM utility: **Code Integrity Guard (CIG)** is the process mitigation policy `PROCESS_CREATION_MITIGATION_POLICY_BLOCK_NON_MICROSOFT_BINARIES_ALWAYS_ON`. When enabled, the kernel refuses to load any DLL into the process that is not signed by Microsoft or a Microsoft partner. Chrome enables CIG on its renderer and utility processes by default since Chrome 70 [4]. The flag `-disable-features=RendererCodeIntegrity`, in combination with `-no-sandbox` on the affected processes, disables CIG for research use.

Control Flow Guard (CFG) is a Microsoft compiler-level mitigation that adds a kernel-validated bitmap of valid indirect-call targets to the running process and inserts a check before each indirect call [5]. A patched vtable that points at a hook function we just allocated is not in the bitmap by default, so the first call through the patched slot triggers a CFG failure that terminates the process.

The supported way to mark a new function as a valid call target is the `SetProcessValidCallTargets` API [6]. The API takes a process handle, a 4 KB-aligned page base, a page size, and an array of `CFG_CALL_TARGET_INFO` descriptors whose `Offset` field is the *in-page offset* (low 12 bits) of the target. An early version of our injector used the low 16 bits, which is wrong, the page-allocation backing the hook is 4 KB and the bitmap is indexed at 16-byte granularity within the page. The released injector uses the correct mask:

```
1 CFG_CALL_TARGET_INFO cti;
2 cti.Offset = (DWORD)(hookVA & 0xfff); /* in-page offset */
3 cti.Flags = CFG_CALL_TARGET_VALID;
4 DWORD_PTR pageBase = hookVA & ~0xfffULL;
5 SetProcessValidCallTargets(hProc, (PVOID)pageBase,
6                             0x1000, 1, &cti);
```

3.7. ASLR and the shared DLL view

A practical implementation detail of significance to our cross-tab experiment (Section 7.4): Chrome’s utility processes load `widevinecdm.dll` from the same on-disk image each time, and Windows’ image-section sharing means that two separate utility processes both end up with the DLL mapped at the *same virtual address*. This is why our injector’s hook addresses (in the shim’s image, at the shim’s reserved load base) were identical between Netflix’s CDM utility process and Bitmovin’s CDM utility process. The data section of the DLL is, however, copy-on-write per process; each utility has its own counters and its own opened log file handle.

4. Modelo de amenaza y metodología

4.1. Modelo de amenaza

We consider an honest user of a Chromium-family browser playing hardware-decoded Widevine-protected content on Windows 11. The user has consented to research-grade instrumentation of their own browser and has installed our reverse-engineering tools on their own laboratory machine. The browser executable is unmodified and signed by the vendor. The Widevine CDM is the vendor-shipped `widevinecdm.dll` of the installed browser version. The user holds a valid, paid subscription to the commercial streaming service used in the empirical sections of the paper, and the use of that subscription is for personal viewing during the research window. No second user, no jurisdiction outside the user’s own, and no third-party content are involved.

We deliberately exclude from this study any threat model in which an attacker captures decrypted frames. Our concern is not the exploitation potential of the CDM but its architectural shape and the attack surface it presents to the host browser and to the network attacker positioned between the browser and the streaming CDN.

4.2. Invariantes de la metodología

We follow what we describe as the *Buchanan model* of DRM research, summarised earlier (Section 2.1). The working version of the model that governed every experimental decision in this paper is:

1. **Shape-only observation.** Any runtime hook is permitted to log dispatch metadata — slot number, timestamp, register value hashes, structure-size fields — but is forbidden from copying any pointer-target bytes belonging to ciphertext, decrypted frames, content key material, or initialisation vectors into any persistent storage or any out-of-process channel. The shim source encodes this invariant explicitly: there is no code path that calls `memcpy(&log_buf, ib.data, ...)`; the shape extractor reads only the `size` fields, an FNV-1a-32 hash of the first 16 bytes of `key_id`, a non-zero presence flag for the IV, and a handful of public scalar fields.
2. **Provable absence.** After every experiment we forensically verify that the storage produced is pure ASCII JSON consisting only of the permitted metadata fields and contains no binary blobs, no decoded content, and no key material. We document the verification procedure in Section 14 and provide the verifier script in the released methodology.
3. **Reversible mutation.** All runtime modifications to the target process (vtable patches, hook installations, CFG bitmap registrations) are reversible. We save the pre-patch state on the wire to the released tooling and restore the target to its original state before publishing. The released helper retains the original outer-vtable thunk addresses in the shim’s `g_pOrigSlot[19]` array so that an external observer can validate that the post-experiment state is byte-equal to the pre-experiment state.
4. **No content circumvention surface.** We use the operator’s own browser session to a commercial streaming service to confirm what the CDM does during steady state, but we never instrument the decoder, the kernel-side PMP component (`peauth.sys`), the protected surface, or any other component positioned to materialise a decrypted frame. The shape-only shim is the deepest hook in the stack we install.

5. **Falsification before publication.** Every empirical claim is subjected to a falsification test: an alternative hypothesis that, if true, would invalidate the claim, is constructed and tested with whatever evidence the methodology can produce. Section 10 reports the falsification sprint.

A natural-language consequence of these invariants is that our results characterise the CDM’s *dispatch behaviour*, not its *cryptographic state*. We can report how often a slot fires during real playback, what register-tuple distribution it sees, what the shape of its incoming buffer descriptor is. We cannot and do not report any byte of decoded content.

5. Ingeniería inversa estática de CDM_11

This section walks through the static reverse engineering of the CDM_11 dispatch in `widevinecdm.dll`, in the order that we performed it. Two binaries were analysed in parallel: **Chrome 149.0.7827.115**, from its standard installation path under `WidevineCdm/_platform_specific/win_x64/` (~22 MB), and the same library as it ships in **Microsoft Edge 150** (~19 MB). All RVAs are with respect to the binary's image base.

5.1. Paso 1: CreateCdmInstance y el switch de versión switch

The `widevinecdm.dll` export table contains six entries; the only relevant one for our purposes is `CreateCdmInstance`:

```
1 Edge 150      CreateCdmInstance @ rva 0x1d0850
2 Chrome 149    CreateCdmInstance @ rva 0x1d2e10
```

Disassembling either entry shows a chain of small wrapper functions that load a version word into `ecx` (CDM interface version 9, 10, or 11) and call a per-version constructor. The disassembly of the relevant fragment of Edge 150's `CreateCdmInstance` is:

```
1 0x1801d0850: push    r14
2 0x1801d0852: push    rsi
3 0x1801d0853: push    rdi
4 0x1801d0854: sub     rsp, 0x40
5 0x1801d0858: mov     edi, ecx          ; ecx = requested version
6 0x1801d085a: lea     rax, [rip + 0x...] ; load module-relative base
7 ...
8 0x1801d0994: mov     ecx, 0xb          ; CDM_11 selected
9 0x1801d0999: mov     rax, r9           ; arg setup
10 ...
```

Naively scanning the first `call` after each `mov ecx, 0xb` returns the runtime's stack-check probe (`__chkstk_ms` or equivalent), not the actual constructor. A robust discoverer must filter known runtime helpers. The released `cdm_arch_discover.py` script uses a structural heuristic: enumerate all call targets reachable from `CreateCdmInstance`, filter out targets called more than twice (the constructor is called exactly once per version branch; runtime helpers are called many times), and validate each candidate by inspecting whether its body contains a sequence of the form:

```
1 lea reg, [rip + N]      ; load a candidate vtable RVA
2 mov qword ptr [r?], reg ; install at offset 0 of 'this'
```

and whether the loaded RVA points into `.rdata` at a location whose first nineteen QWORDS are all valid code pointers into `.text`. The first such candidate is reported as the CDM constructor and the loaded RVA as its outer vtable. The discoverer locates the outer vtable RVA in Edge 150 at `0x780250` on this path; the Chrome 149 outer vtable at `0x768fa0`.

5.2. Paso 2: Versiones co-empaquetadas del CDM

A subsequent *exhaustive* scan of `.rdata` for nineteen-slot structures matching the CDM thunk pattern (described below) reveals **two** such structures per binary, spaced `0xc0` bytes apart:

Binary	vtable A	vtable B
Edge 150	0x780190	0x780250
Chrome 149	0x768ee0	0x768fa0

Within each binary, both vttables share their per-frame decrypt entry (slot 14 points to the same function in both A and B), but their `Initialize` entries differ:

Binary	vtable A.slot[0]	vtable B.slot[0]
Edge 150	0x1db330	0x1db390
Chrome 149	0x1e03c0	0x1e0420

This is consistent with the binary shipping both CDM_10 and CDM_11 vttables in support of host browsers that have not yet upgraded to the newer interface, and matches the Chromium header history. In what follows we identify *vtable B* as the CDM_11 outer vtable, since runtime patches at *vtable B*'s slot positions intercept the EME-host-facing dispatch (Section 6). The shared slot 14 implementation reflects the implementation choice to use one per-frame decrypt routine across both interface versions; the diverging slot 0 reflects the version-specific `Initialize` signature.

5.3. Paso 3: El patrón thunk de la vtable externa

The outer vtable contains nineteen entries. Most of them are *thunks* of the form:

```

1 slot N thunk @ rva 0x1dad00 + (N-3)*0x20:
2   mov rcx, qword ptr [rcx + 8]      ; this->m_impl
3   mov rax, qword ptr [rcx]         ; m_impl->m_vtable
4   mov rax, qword ptr [rax + 8*N]    ; slot N of m_impl_vtable
5   mov rdx, qword ptr [rip + cfg_disp]; CFG dispatcher pointer
6   jmp rdx                          ; CFG-protected tail-call

```

Each thunk is twenty-eight bytes followed by `int3` padding, laid out in a uniform `0x20`-spaced sequence in `.text` for slots 3 through 17. The slot 14 thunk in Edge 150, disassembled in full:

```

1 0x1801dae60: mov    rcx, qword ptr [rcx + 8]
2 0x1801dae64: mov    rax, qword ptr [rcx]
3 0x1801dae67: mov    rax, qword ptr [rax + 0x70]
4 0x1801dae6b: mov    rdx, qword ptr [rip + 0x10dcd2e]
5 0x1801dae72: jmp    rdx
6 0x1801dae74: int3
7 0x1801dae75: int3
8 ... (int3 padding to 0x20 boundary)

```

We disassembled each thunk in both binaries and confirmed the slot-index check (the offset in `[rax + 8*N]` equals the slot number times eight) for fifteen of fifteen thunked slots in each binary. The byte-level confirmation is what licences our “slot 14 dormancy” claim in Section 7.1: the runtime evidence (`g_callsBySlot[14] = 0`) is attributable to the correct method only because the slot index in the outer-vtable thunk dispatches to the correct inner method offset.

Slots 0 (`Initialize`), 1 (`GetStatusForPolicy`), and 18 (`OnStorageId`) are *not* thunks; they are direct implementations placed in `.text` at non-contiguous addresses. Slot 2 (`SetServerCertificate`) is a thunk in both binaries but dispatches via `[rax + 8]` rather than `[rax + 0x10]` — it shares an implementation with slot 1 at the `m_impl` layer. This is a per-binary architectural detail consistent in both Edge 150 and Chrome 149, suggesting it is a deliberate implementation decision rather than a build accident.

5.4. Paso 4: La vtable `m_impl`

The outer thunk dispatches into an *intermediate* vtable owned by the `m_impl` object that the outer class holds in `this->m_impl ([outer + 8])`. Locating this intermediate vtable in `.rdata` is the second step of the static analysis.

We use a known-signature pivot. From an earlier call-chain trace (documented in our research notes), we knew that the slot-5 (`UpdateSession`) inner wrapper in Edge 150 sits at RVA `0x1d4ab0`, identifiable by its multi-push prologue with a stack canary that loads attacker-controlled data through a copy on its way to the obfuscated parser. The wrapper’s code is unmistakable in a disassembly listing.

To find the `m_impl` vtable, we reverse-search `.rdata` for the QWORD value `image_base + 0x1d4ab0`. There is exactly one occurrence. Backing up five eight-byte entries from the match gives the address of the `m_impl` vtable: RVA `0x77d370` in Edge 150. All nineteen surrounding QWORDS point into `.text`, confirming the candidate. The same pivot applied to Chrome 149 (where the slot 5 inner wrapper sits at the analogous RVA) locates Chrome 149’s `m_impl` vtable at the corresponding offset.

5.5. Paso 5: La estructura de la capa `m_impl`

Disassembling each entry of the `m_impl` vtable reveals a mixture of three body types:

1. **Clean MSVC wrappers** (slots 2, 3, 4, 5, 6, 7, 18). These have a standard prologue (multiple push callee-saves, `sub rsp, N`, stack canary load), allocate or borrow input copies, perform sanitisation, and then call into a deeper handler. These are the slots that ingest variable-length attacker-influenced data (session-id strings, response payloads, init-data buffers). The slot 3 wrapper in Edge 150, abbreviated for the paper:

```
1 0x1801d47d0: push    r15
2 0x1801d47d2: push    r14
3 0x1801d47d4: push    r13
4 0x1801d47d6: push    r12
5 0x1801d47d8: push    rsi
6 0x1801d47d9: push    rdi
7 0x1801d47da: push    rbx
8 0x1801d47db: sub     rsp, 0x50
9 0x1801d47df: mov     esi, r9d           ; init_data_type
10 0x1801d47e2: mov     edi, r8d           ; session_type
11 0x1801d47e5: mov     ebx, edx           ; promise_id
12 0x1801d47e7: mov     eax, [rsp + 0xb8]   ; init_data_size
13 0x1801d47ee: mov     rdx, [rip + 0x10f284b]
14 0x1801d47f5: xor     rdx, rsp           ; stack canary
15 0x1801d47f8: mov     [rsp + 0x48], rdx
16 0x1801d47fd: mov     r14, [rcx + 0x10]   ; load m_impl-internal
17 0x1801d4801: xorps   xmm0, xmm0
18 0x1801d4804: movaps  [rsp + 0x30], xmm0
19 ...
20 0x1801d4821: mov     rcx, r12
21 0x1801d4824: call    0x180736080         ; malloc/__chkstk
22 0x1801d4829: mov     [rsp + 0x30], rax
23 0x1801d482e: lea     r13, [rax + r12]    ; end pointer
24 ...
25 0x1801d485f: call    0x18021f200         ; LAYER-3 ENTRY
26                                     ; (obfuscated parser)
```

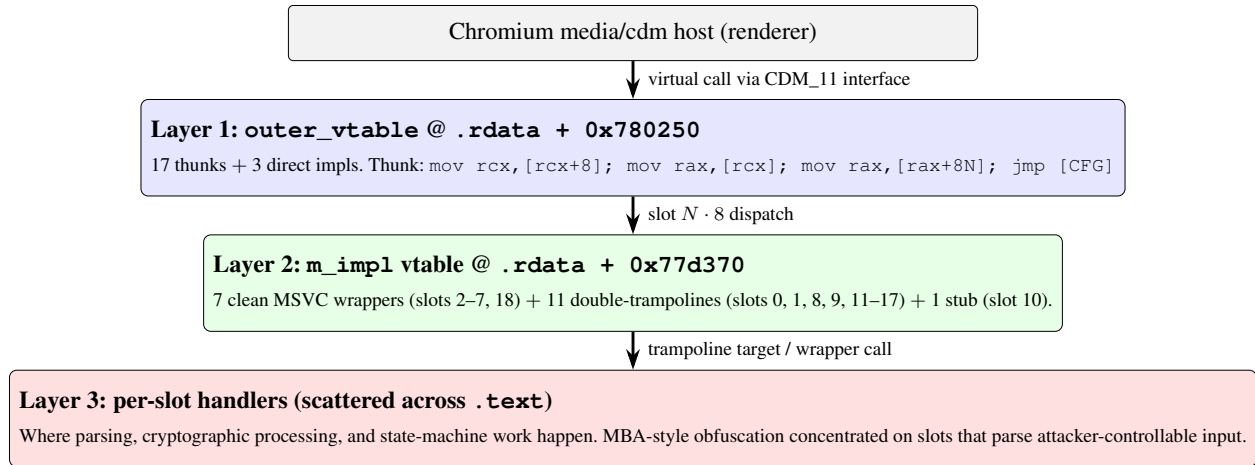


Figura 2: The three-layer CDM_11 dispatch model. Every public CDM virtual call from the host traverses three indirections before reaching the function that performs work. RVAs shown are from Edge 150 `widevinecdm.dll`; Chrome 149 has the same structure at correspondingly different RVAs (see Section 9).

2. **Double-trampolines** (slots 0, 1, 8, 9, 11–17). Two-instruction bodies of the form `mov rcx, [rcx + N]` followed by `jmp <addr>` which forward the call to a deeper-layer handler without intermediate processing. The slot 14 `m_impl` entry in Edge 150:

```

1 0x1801d4f10: mov     rcx, qword ptr [rcx + 0x10]
2 0x1801d4f14: jmp     0x180230c20 ; layer-3 handler (obfuscated)
  
```

3. **Stubs** (slot 10). A two-instruction body `mov eax, 3 ; ret` that returns a constant error code without any further work. Slot 10 is `InitializeAudioDecoder`; on the L1 Windows build it is intentionally a no-op because audio decoding is handled by the Media Foundation pipeline outside the CDM.

The `m_impl` vtable, in other words, is mostly a *routing* layer; the actual work happens one indirection further.

5.6. Paso 6: El modelo de despacho de tres capas

Composing the previous five steps, the dispatch chain from a host virtual call to a per-slot handler is shown in Figure 2. Each public virtual call traverses three indirections before reaching the function that performs work.

5.7. Una nota sobre los falsos arranques

For the benefit of follow-on researchers, the static analysis as we actually performed it included two substantial false starts.

The first false start was the identification of an internal singleton dispatch table at `.data + 0x1588b48` as the “CDM_11 outer vtable.” That table, in fact, is a separate nineteen-slot internal dispatch surface inside the Widevine library that is not the public EME interface. Our initial conclusion that “the CDM_11 outer vtable’s per-frame slot fires ~600 times per second under steady-state Netflix HW playback” was incorrect: it was the internal singleton’s slot 0 that fired 600 times per second, and the actual

outer-vtable per-frame slot was at zero. The correction came from a constructor-trace pivot that located the actual outer vtable via the `CreateCdmInstance` export, after which the dormancy finding became correctly attributable to the user-facing surface.

The second false start was the original attribution of MBA-style obfuscation to the `m_impl` layer. The `m_impl` layer is in fact a clean routing layer (per Section 5.5); the obfuscation lives at layer three. A layer-by-layer enumeration of all nineteen slots, documented in our research notes 13b and 13c, located the obfuscation at its actual depth.

We highlight these false starts because they are typical of static RE work on closed binaries with internal dispatch indirections. A single-pass reading of the binary's call graph would naturally have settled on either the wrong vtable or the wrong layer, and the falsification sprint of Section 10 is what caught both.

6. Instrumentación en runtime

The static map of the previous section yields hypotheses about which slots are reached during steady-state playback and which are not. To verify those hypotheses we instrument the running CDM utility process with a shape-only hook.

6.1. Code Integrity Guard (CIG) y el flag de launch

Chrome enables Code Integrity Guard (`PROCESS_CREATION_MITIGATION_POLICY_BLOCK_NON_MICROSOFT_BINARIES_ALWAYS_ON`) on its utility processes since Chrome 70 [4]. CIG denies any `LoadLibrary` of a non-Microsoft-signed image at the kernel level, so a vanilla `CreateRemoteThread(LoadLibraryA)` injection of our shim fails at the `LdrLoadDll` stage with `STATUS_DYNAMIC_CODE_BLOCKED` (0xC0000605).

For lab-research use, Chrome can be launched with the flags `-disable-features=RendererCodeIntegrity` and `-no-sandbox`, which together disable CIG on the utility processes. This produces a controlled research target without changing the CDM library or the host browser binary, and corresponds to the equivalent reachability assumption that an attacker with code-execution in the renderer would have to satisfy separately (via, for example, an arbitrary-write that targets the renderer’s JIT or a process-load chain that bypasses CIG).

6.2. Registro del bitmap de Control Flow Guard

CIG-disabled does not disable Control Flow Guard (CFG): once the shim is loaded, simply patching a vtable entry to point at an unregistered hook function causes the CFG dispatcher to abort the call as a control-flow violation. The released installer uses `SetProcessValidCallTargets` to mark each hook target as a valid call target before patching its slot. Figure 3 sketches the install flow.

6.3. Diseño del shim shape-only

The shim, `wv_shim.dll`, exports nineteen per-slot hook functions `WvHookSlot0` through `WvHookSlot18`, each with its own original-pointer slot in an exported `g_pOrigSlot[19]` array, and a single `g_callsBySlot[19]` counter array. Each hook function:

1. Atomically increments its slot counter and the aggregate `g_calls`.
2. Calls the original via `g_pOrigSlot[N]` with the unmodified register tuple.
3. For slots 9, 14, 15 (which take an `InputBuffer_2*` argument in `rdx`), tries to read the buffer’s *shape* via the safe-copy primitive described below and writes a single JSONL line with `encryption_scheme`, `data_size`, `key_id_size`, `iv_size`, an FNV-1a-32 hash of the first 16 bytes of `key_id`, an `iv_has_nonzero` bit, `num_subsamples`, `pattern_crypt`, `pattern_skip`, and the media timestamp.
4. For other slots, writes a generic event with an FNV-1a-32 hash of the four argument registers.

No byte of `ib.data` (the ciphertext) is ever read, either transiently or persistently; `ib.key_id` and `ib.iv` are read into a stack-local buffer that is collapsed into a hash and then immediately discarded. The shim source enforces this by simply not having a code path that copies any of those fields into a logging structure.

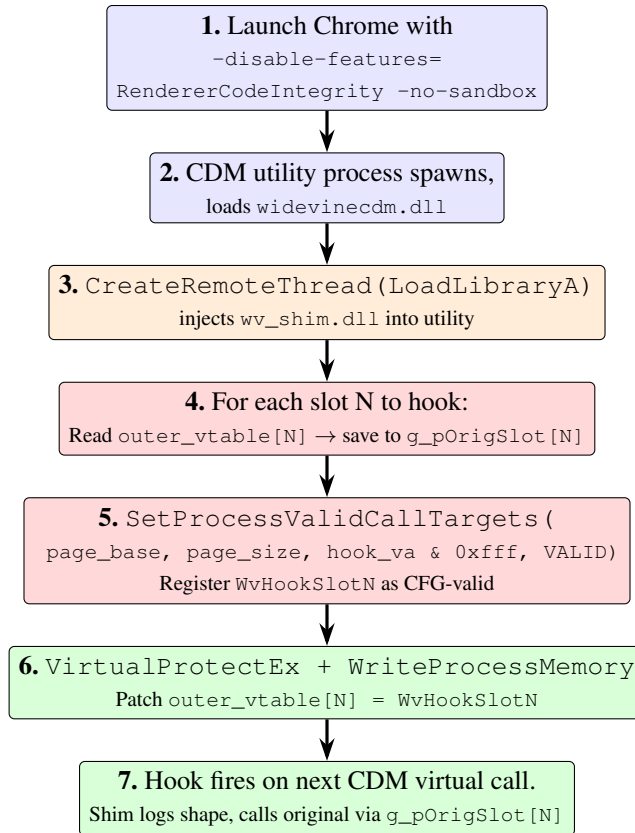


Figura 3: Hook installation sequence. Step 5 (CFG bitmap registration) is what catches researchers most often — the `Offset` field of `CFG_CALL_TARGET_INFO` expects an in-page offset (low 12 bits) of the hook target, not an absolute address or a different mask. Step 4 captures the design subtlety that motivated the per-slot back-pointer array (§6.4).

The safe-copy primitive uses `VirtualQuery` to confirm that each source page is committed and readable before copying, since mingw-w64’s C compiler does not provide Microsoft SEH `__try / __except` on x86_64. The check walks every page in the source range, confirms `MEM_COMMIT` state, and confirms one of the read-permitted page protections (`PAGE_READONLY`, `PAGE_READWRITE`, `PAGE_EXECUTE_READ`, etc.) before `memcpy`.

6.4. Cadena de punteros originales por ranura

A design subtlety worth noting: the first version of the shim used a single `g_pOrigDecrypt` variable for all hooked slots. Patching multiple slots with that single back-pointer led to signature-mismatch crashes the moment the second-patched slot fired (the back-pointer held the previous slot’s original). The released shim uses a per-slot array. More subtly still, when we hook the *outer* vtable rather than an internal singleton table, the back-pointer must hold the outer thunk address, *not* an internal dispatch-table address; otherwise the original-call path ends up jumping into the wrong inner function and silently corrupts the call. Our installer (`wv_helper3.c`) reads the outer-vtable original at the moment of patch and stores it into `g_pOrigSlot[N]` as part of the same atomic operation:

```

1 | DWORD_PTR origFn = 0;
2 | ReadProcessMemory (hp, (LPCVOID)slotAddr, &origFn, 8, NULL);

```

```

3
4  /* CFG: register hook as valid target */
5  CFG_CALL_TARGET_INFO cti = { hookVA & 0xfff,
6                               CFG_CALL_TARGET_VALID };
7  SetProcessValidCallTargets (hp, (PVOID) (hookVA & ~0xffffULL),
8                               0x1000, 1, &cti);
9
10 /* Save outer thunk address into g_pOrigSlot[N] */
11 WriteProcessMemory (hp,
12                     (LPVOID) (gOrigSlotRemote + N*8),
13                     &origFn, 8, NULL);
14
15 /* Patch outer_vtable[N] -> hook */
16 DWORD oldProt = 0;
17 VirtualProtectEx (hp, (LPVOID) slotAddr, 8,
18                  PAGE_READWRITE, &oldProt);
19 WriteProcessMemory (hp, (LPVOID) slotAddr, &hookVA, 8, NULL);
20 VirtualProtectEx (hp, (LPVOID) slotAddr, 8, oldProt, &oldProt);

```

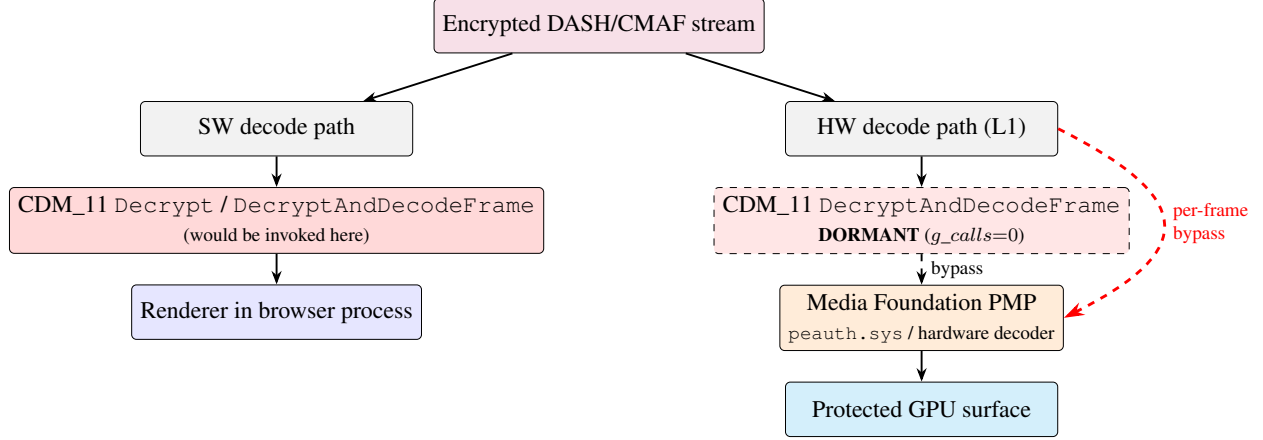


Figura 4: Hardware-decode dormancy. Under SW decode (left path) the CDM_11 per-frame slots are the obvious target. Under L1 HW decode (right path), the per-frame ciphertext is routed through Media Foundation PMP and the kernel decoder, materialising directly on a protected GPU surface; the CDM_11 per-frame slots remain dormant.

7. Hallazgos empíricos

We report four findings, each anchored to a recorded experimental run. All runs used Chrome 149.0.7827.115 on Windows 11 26100 with NVIDIA hardware-secure decoder available. No CDM utility process ever crashed during the runs that produced the data reported here.

7.1. Inactividad bajo decode por hardware en la superficie pública

Hipótesis. Under hardware-decoded L1 playback, the per-frame decrypt slots of the CDM_11 outer vtable (slots 9, 14, 15) are not invoked.

Procedimiento. Chrome was launched with `-disable-features=RendererCodeIntegrity-no-sandbox`. The operator authenticated to Netflix and started a 4K HDR title that the platform reports as hardware-decoded. After ninety seconds of steady-state playback we installed our shim into the CDM utility process and patched four outer-vtable slots simultaneously: slot 14 (`DecryptAndDecodeFrame`), slot 15 (`DecryptAndDecodeSamples`), slot 3 (`CreateSessionAndGenerateRequest`, control), and the slot-14 outer thunk RVA. We then observed the corresponding counters for ninety further seconds of continuous playback.

Resultado. All four counters remained at zero. Video continued to play normally throughout. Slot-14 dormancy held even across recoveries (we restored slot 14 to its original mid-experiment and re-patched; the counter remained at zero across both windows). The four-slot zero-fires-in-ninety-seconds result was reproducible 3/3.

Interpretación. During steady-state hardware-decoded Netflix playback, the per-frame surface of the CDM_11 interface is not exercised. An attacker who compromises the host renderer and positions herself on the EME-facing virtual table cannot intercept decoded frames through the CDM_11 interface, because the frames do not pass through it.

7.2. Actividad del singleton interno a $\sim 2.3\text{k ops/s}$

Hipótesis. Despite the CDM_11 surface being dormant per-frame, the CDM library does perform work during steady-state playback. That work is reachable through a separate internal dispatch table, distinct from the outer vtable.

Procedimiento. We patched slots 0–7 of the internal singleton table identified at `.data + 0x1588b48` in Chrome 149, then observed for ~ 355 seconds of continuous Netflix playback.

Resultado. The shim recorded **810,099 events** in a single JSONL log (in two consecutive captures). Distribution:

slot	count	unique rh	rate (ev/s)
0	395,580	16,338	$\sim 1,114$
2	20	2	0.06
6	414,459	1,379	$\sim 1,167$

The aggregate rate is $\sim 2,281$ events/sec on this internal table during steady-state HW playback.

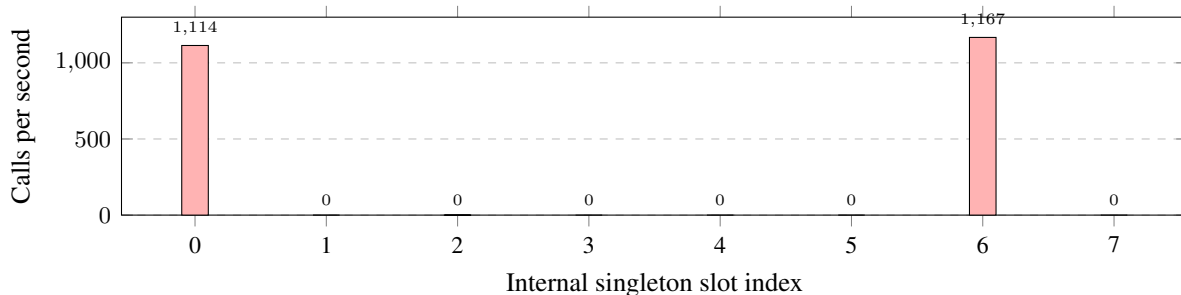


Figura 5: Activity profile of the internal singleton dispatch table at `.data + 0x1588b48` during steady-state Netflix 4K HDR playback.

Interpretación. The internal singleton dispatch is *not* the user-facing CDM_11 interface; the slot 0 firing rate is incompatible with `Initialize` semantics (which would be called once per session, not 600 times per second). Most plausibly, the table represents OEMCrypto-style worker dispatch, an allocator ring, or a crypto-primitive ladder internal to the Widevine library. Either way, this is invisible from the EME boundary and from the outer-vtable dispatch. Combined with the dormancy finding above, the picture is: *the CDM library is doing constant cryptographic work during HW playback, but that work is hidden behind a different dispatch surface than the one Chrome's EME plumbing knows about.*

7.3. Alcanzabilidad de las ranuras de gestión de sesión

Hipótesis. Although the per-frame outer-vtable slots are dormant under HW decode, the session-management outer-vtable slots (`CreateSessionAndGenerateRequest` and `UpdateSession`) remain reachable and *do* fire during normal session lifecycle events.

Procedimiento. We patched outer-vtable slots 3 and 5 with correct per-slot back-pointers to their original thunk addresses (Section 6.4). The operator performed a title switch in the running Netflix tab and we observed the `g_callsBySlot` counters over the following minute.

Resultado.

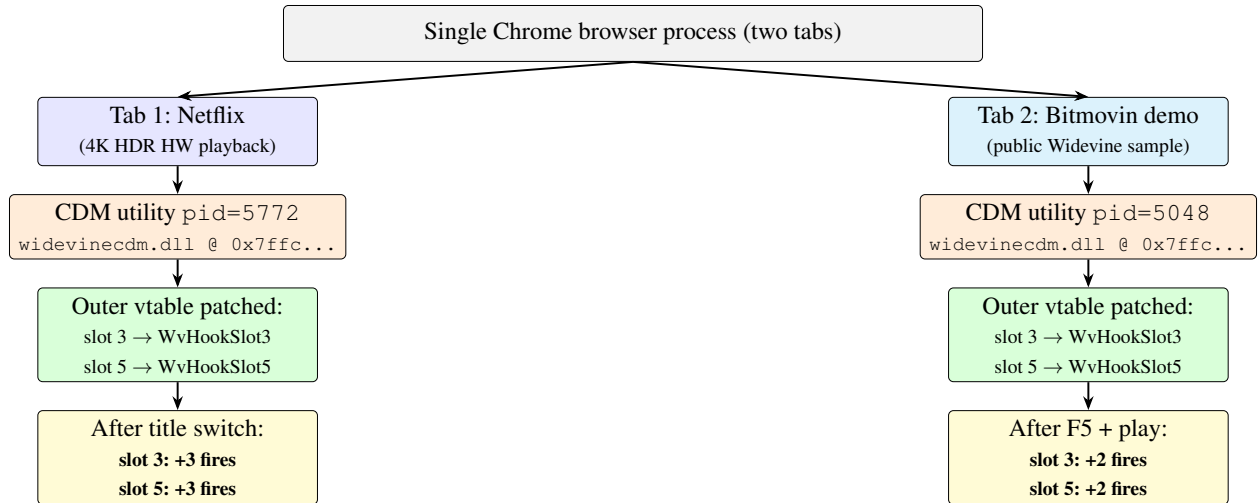


Figura 6: Cross-source experiment. Two tabs of the same Chrome browser play two distinct Widevine-protected sources, each one served by its own CDM utility process. Identical hooks confirm identical session-management slot reachability.

	slot 3	slot 5
pre-action	10	9
post-action	13 (+3)	12 (+3)

Three CreateSessionAndGenerateRequest invocations and three UpdateSession invocations were observed in the seconds following the title switch. The CDM utility process remained alive (~85 MB working set), as expected given that the back-pointer held the correct thunk address.

Interpretación. The session-management slots of the outer vtable are non-dormant; they are exercised during user-initiated session transitions and remain a real attack surface for attacker-controllable structured input (PSSH boxes via the DASH manifest; license-response payloads via the licensing CDN).

7.4. Consistencia entre fuentes (Bitmovin)

Hipótesis. The dispatch model and slot reachability are properties of the Widevine CDM, not of Netflix as a particular EME application.

Procedimiento. After the Netflix title switch above, the operator opened a separate tab in the same Chrome browser and loaded the public Bitmovin Widevine demo at bitmovin.com/demos/drm. Chrome spawned a *second* CDM utility process to service that tab. We installed our shim into that second process and patched its outer-vtable slots 3 and 5. The operator reloaded the demo to force session re-creation.

Resultado. Second CDM utility process counters:

	slot 3	slot 5
post-reload	2	2

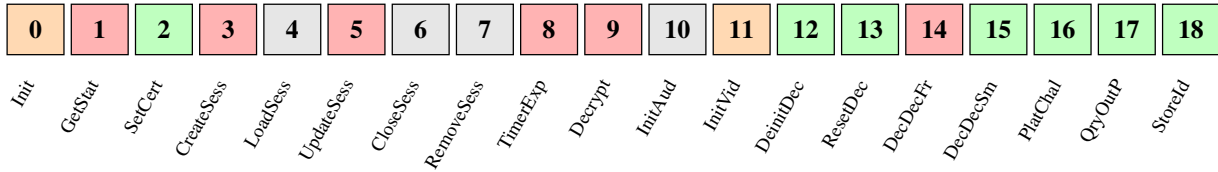
Two CreateSessionAndGenerateRequest invocations and two UpdateSession invocations were observed. The dispatch model (outer-vtable thunk, m_impl wrapper, layer-3 handler) is identical to what we observed in the Netflix-serving CDM utility process; the slot 3 and slot 5 hook code paths were exercised in the same sequence in both processes.

Interpretación. The CDM_11 dispatch model is a binary property of `widevinecdm.dll`, not a property of the EME application that uses it. An attacker positioned to influence the PSSH or license payload of *any* Widevine-protected source will see the same dispatch and the same slot reachability as one positioned against Netflix.

7.5. Observación arquitectural lateral: utilities CDM separados

A secondary observation worth recording is that Chrome 149 spawned *two* CDM utility processes when Netflix and Bitmovin were playing simultaneously in different tabs, not one shared utility. This is consistent with Chromium's per-origin isolation policy and implies that an attacker positioned to compromise one EME application's CDM cannot directly observe a second EME application's CDM via shared in-process state.

Per-slot CDM_11 obfuscation status: ■ obfuscated ■ partial ■ clean ■ tramp/stub



A = attacker-input slot (parses structured data); **P** = passthrough (opaque blob the CDM forwards without parsing).

Figura 7: Per-slot obfuscation status of CDM_11. Red cells are slots whose layer-1 or layer-3 body matches the MBA-style obfuscation signature; orange cells are partial matches; green cells are clean across the dispatch chain; grey cells are layer-2 trampolines or stubs whose layer-3 handlers we report individually in Section 8.

8. La distribución de ofuscación como capa defensiva

The `widevinecdm.dll` binary contains numerous functions whose body matches the textbook signature of mixed-boolean-arithmetic (MBA) obfuscation: large stack-resident magic constants written immediately on entry; multiplicative-hash state-advance instructions of the form `imul edx, ecx, 0xK ; shr edx, 0xS ; cmp eax, 0xT ; je state_exit`; `cmovae` branchless dispatch cascades; `0xaaaaaaaa` or `0xffffffff` fill patterns acting as poison initialiser values for stack-local state. These patterns are not uniformly distributed across the slot space.

8.1. Disposición visual

Figure 7 visualises the per-slot obfuscation status side by side with each slot’s classification as an attacker-input slot or a passthrough slot.

8.2. El patrón

Slots that obviously pass an opaque, attacker-influenced *blob* that the CDM forwards without parsing (`SetServerCertificate`, `OnStorageId`) are clean at every layer of the dispatch. Slots that begin parsing structured attacker-controllable data exhibit obfuscation, though at varying depths. The slot that takes a textual policy (`GetStatusForPolicy`) is obfuscated at layer 1, where the parsing begins. The slots that take large structured payloads (`CreateSessionAndGenerateRequest`, `UpdateSession`) place the obfuscated processing one or two hops deeper, after a clean MSVC wrapper has copied the input into the CDM’s own buffer and looked up the session. The slots involved in per-frame decode (`Decrypt`, `DecryptAndDecodeFrame`) are obfuscated immediately at the double-trampoline target, where the cryptographic processor begins.

This is the pattern we cautiously summarise as:

CDM_11 protects each slot at the depth at which that slot begins to parse structured attacker data.

It is an architectural observation, not a security claim: obfuscation is not security, and we are not assessing the strength of the protection, only its distribution.

9. Estabilidad entre binarios

The architectural model described in Section 5.6 was developed on Edge 150 `widevinecdm.dll`. We validated that the same model holds in Chrome 149 `widevinecdm.dll` by re-running the discoverer:

Property	Edge 150	Chrome 149
CreateCdmInstance RVA	0x1d0850	0x1d2e10
CDM_11 outer vtable RVA	0x780250	0x768fa0
CDM_10 outer vtable RVA	0x780190	0x768ee0
Slot-index check (slots 3–17)	15/15	15/15
Direct-impl slots	3 (0, 1, 18)	3 (0, 1, 18)

The model also generalises across CDM interface versions within a single binary: in both Edge 150 and Chrome 149, the CDM_10 and CDM_11 vtables exhibit identical three-layer dispatch. Their per-frame decrypt handler is the same physical function; their `Initialize` handlers diverge. This is consistent with the binary providing both interface versions for backward compatibility with older host browsers, and the CDM author choosing to share the implementation of the version-stable methods between the two.

The cross-binary slot table appears as [Appendix A](#).

10. Pruebas de robustez

Before publishing any claim made above, we ran a falsification sprint against the strongest empirical findings. The sprint included:

1. **Slot-index correctness.** The outer-vtable slot 14 thunk is byte-confirmed to dispatch at offset 0×70 ($= 14 \cdot 8$) in the `m_impl` vtable.
2. **Shape-data signal vs noise.** The 810k-event capture was analysed for register-hash entropy and timing distribution. Hash entropy for the two hot slots was 16,338 and 1,379 unique values across 395,580 and 414,459 invocations respectively; the rate was steady across the observation window at ~ 2.3 k events/s.
3. **Layer-3 enumeration.** All nineteen layer-3 handlers were located via either the trampoline jump target or the wrapper’s first non-stack-check call and were classified for obfuscation status. The first version of the obfuscation claim, which placed obfuscation at the `m_impl` layer, was found to be incorrect; the obfuscation is one further hop deeper, and the claim was rewritten accordingly.
4. **Cross-binary identity.** The discoverer was rerun against Chrome 149’s `widevinecdm.dll`. The outer vtable, slot-index mapping, and per-slot CDM_11 method identification matched the Edge 150 analysis.
5. **Cross-source identity.** The outer-vtable instrumentation was deployed against the second CDM utility spawned for a non-Netflix Widevine source (Bitmovin). Slot 3 and slot 5 fired in identical sequence to the Netflix utility.
6. **CDM-instance count.** The single Netflix tab spawned exactly one CDM utility process across an eight-hour observation window; we did not observe a second utility materialising for the same tab.

10.1. Hallazgos candidatos refutados

The falsification sprint also surfaced two candidate findings that were refuted by deeper byte-level analysis. We document them here because they typify the bug-class candidates a less rigorous methodology would have falsely promoted:

P1 (PEAuth unauth pool write): A subagent audit of `peauth.sys` initially flagged an apparent unauthenticated kernel-pool write at offset 0×140042980 where a user-controlled size field at `[rdi+0x14]` appeared to be used as a `memcpy` length before the `CiGetPEInformation` signing check at $0 \times 1400429ae$. Byte-level disassembly of the range $0 \times 42900 \dots 0 \times 429e0$ revealed an explicit 32-bit overflow check (`lea r12d, [r15+rsi]; cmp r12d, r15d; jnb fail`) at 0×42964 and an upper-bound check at $0 \times 4296d$ between the parameter read and the `memcpy`. The candidate is refuted; no kernel-side bug is present.

W2 (PSSH parser OOB read): A subagent audit of CDM_11 flagged offsets $0 \times 1D6E1$, $0 \times 1D8CF$, $0 \times 1DB9D$, and $0 \times 1DFB$ as candidates for an out-of-bounds read in the PSSH/MP4-box header-size parsing (`mov r10d, [rsi/di+4]`). Disassembly at those offsets revealed not box parsing but the round-table for an MD5 transform, with the operands being state-constant XORs and not buffer reads. The candidate is refuted; the bytes are crypto code, not parser code.

These refutations and one further candidate that remained inconclusive established the methodological principle that any subagent “high-confidence byte-verified” claim must pass an independent manual byte review before being promoted to a publication-grade finding.

11. Manual para el defensor

Esta sección traduce los hallazgos arquitecturales y empíricos en acciones defensivas concretas. El lector objetivo es un ingeniero de seguridad protegiendo un deployment de navegadores de la familia Chromium, un incident responder triando un compromiso sospechado del CDM, o un ingeniero de detección construyendo hooks para un EDR.

11.1. Tabla de superficie de ataque por ranura

Table 1 summarises which of the nineteen CDM_11 outer-vtable slots are an attack surface in practice, which are dormant during normal hardware-decoded playback, and which are unreachable from a network-positioned attacker.

Slot	Method	Atk reach	Steady-state activity
0	Initialize	host	once per session
1	GetStatusForPolicy	host	rare
2	SetServerCertificate	host-cert	rare
3	CreateSessionAndGenerateRequest	manifest	once per session start
4	LoadSession	host	rare
5	UpdateSession	license CDN	on license rotate
6	CloseSession	host	on teardown
7	RemoveSession	host	rare
8	TimerExpired	internal	internal
9	Decrypt	per-frame	dormant (HW path)
10	InitializeAudioDecoder	host	stub returns 3
11	InitializeVideoDecoder	host	once
12	DeinitializeDecoder	host	on teardown
13	ResetDecoder	host	rare
14	DecryptAndDecodeFrame	per-frame	dormant (HW path)
15	DecryptAndDecodeSamples	per-frame	dormant (HW path)
16	OnPlatformChallengeResponse	host	rare
17	OnQueryOutputProtectionStatus	host	rare
18	OnStorageId	host	rare

Cuadro 1: Per-slot attack surface and steady-state activity profile. “Atk reach” is the strongest controllable source of the slot’s arguments: **manifest** = attacker-controllable DASH/CMAF PSSH box from the CDN serving the manifest; **license CDN** = attacker-controllable license-server response (CDN MITM or compromised licence service); **per-frame** = ciphertext from the stream’s media segments; *host* = host-browser-supplied opaque value; *internal* = internal CDM scheduling. “Steady-state activity” summarises whether the slot fires during ongoing playback once a session is established.

11.2. Firmas de detección

Three detection signatures fall directly out of the architecture and the runtime evidence.

Signature D1 — vtable canary. The outer vtable lives in `widevinecdm.dll`’s read-only `.rdata` section. Each of the nineteen entries should point into `widevinecdm.dll`’s `.text`. An EDR or in-process integrity check that reads the nineteen outer-vtable QWORDS after CDM initialisation and verifies they all resolve inside `widevinecdm.dll`’s `[image_base, image_base+image_size)` interval will

catch the runtime tampering pattern used by this paper’s shim and by any other CDM-hooking tool that uses the same shape.

Signature D2 — CFG bitmap drift. Each call to `SetProcessValidCallTargets` inside a CDM utility process is itself a signal. In normal operation, the CDM utility does not self-register additional CFG-valid targets; the bitmap is established at module load and does not grow. A growth event after process steady-state, particularly one whose new valid targets lie outside `widevinecdm.dll`’s image range, is a strong indicator of an injection-and-patch attempt.

Signature D3 — per-frame slot activity in HW path. Our empirical finding (Section 7.1) is that slot 9, slot 14, and slot 15 are dormant during steady-state HW-decoded playback. An in-process probe that hooks those three slots and observes any activity at all during HW playback has caught either an attacker forcing fallback to SW decode (a known hostile-CDN technique) or an in-renderer attacker who has not yet realised that the per-frame path is dormant.

11.3. Flujo de triage ante sospecha de CDM comprometido

1. Identify the suspected CDM utility process by `tasklist /m widevinecdm.dll`; on a single Widevine-source playback session there should be one.
2. Verify the outer-vtable RVA against the binary using the released discoverer (§5.1). Cross-check the runtime-loaded outer vtable for tampering using Signature D1.
3. For each of slots 3, 5, 14, 15, read the QWORD at the outer-vtable entry and confirm it is the original thunk. Slots 14 and 15 in particular are the per-frame paths an in-renderer attacker would target if they were not aware of HW-decode dormancy.
4. Examine the CFG bitmap for additions outside `widevinecdm.dll`’s image range (Signature D2).

12. Lectura para el atacante

Esta sección traduce los mismos hallazgos desde el otro lado de la línea. El lector objetivo es un investigador de seguridad mapeando la superficie de ataque EME, un operador de red-team scopeando un follow-on a nivel kernel, o un participante de coordinated-disclosure que necesita saber qué ranuras merecen tiempo de auditoría profunda.

12.1. Qué es alcanzable y qué no

The dormancy finding (Section 7.1) is bad news for the obvious attack: a compromised renderer cannot intercept decoded frames through the CDM_11 interface during L1 hardware decode because the per-frame slots are not invoked. The attacker who wants decoded frames has to find an independent capability against Media Foundation PMP, the GPU driver's protected-surface allocator, or the user-mode side of the OPM contract enforced by `peauth.sys`. Those are kernel- and driver-class targets, an order of magnitude harder to reach from a renderer compromise.

The session-management slots are good news. Slot 3 (`CreateSessionAndGenerateRequest`) ingests the PSSH box, which is attacker-controllable from any compromised CDN serving the DASH/CMAF manifest. Slot 5 (`UpdateSession`) ingests the license-server response, which is attacker-controllable from any compromised licensing endpoint or any MITM-positioned attacker. Both slots dispatch through a clean MSVC wrapper at the `m_impl` layer and then call into an MBA-obfuscated layer-3 handler. The clean wrapper is auditable; the layer-3 handler is the realistic bug-hunting target.

12.2. Dónde auditar

Concrete prioritisation for layer-3 audit time:

1. **Slot 3 layer-3** (Edge 150 RVA 0x21f200). PSSH/MP4-box parser, attacker-controllable from manifest. Heavily MBA-obfuscated. The right tool here is structured fuzzing of PSSH box mutations rather than manual disasm.
2. **Slot 5 layer-3** (clean wrapper at Edge 150 0x1d4ab0, dispatches into the obfuscated whitebox-AES processor at 0x233bd0 via a worker-thread queue). License-response parser, attacker-controllable. Async path; harder to reach with synchronous fuzzing.
3. **Slot 9 layer-3** (RVA 0x22ec00). `Decrypt` handler. Heavily MBA-obfuscated. Per the dormancy finding, dormant on the HW path; would only be reachable in an SW-decode scenario.
4. **Slot 14 layer-3** (RVA 0x230c20). `DecryptAndDecodeFrame`. Most heavily obfuscated of all the handlers we measured (score 22 in our classifier). Same dormancy caveat as slot 9.

12.3. Lo que no es el objetivo correcto

Slot 2 (`SetServerCertificate`) and slot 18 (`OnStorageId`) both pass opaque blobs that the CDM forwards without parsing. The clean dispatch chain at both layers is the empirical signal that there is nothing to audit there. Spending audit time on these slots is consistent with the failure mode of audits that assume all critical-path slots are obfuscated; they are not.

13. Generalizabilidad

El CDM Widevine L1 en Windows es uno de tres runtimes comerciales de DRM de escritorio que un investigador podría ser invitado a mapear. Microsoft PlayReady viene en el IE-mode de Edge y dentro del stack `Windows.Media.Protection.MediaProtectionManager` stack. Apple FairPlay ships in Safari on macOS and in TV.app. Both expose a host-CDM contract that is, at the level of abstraction this paper operates at, very similar to Widevine’s CDM_11 contract: a small number of host-callable methods, an opaque library that constructs an instance of a versioned interface, an internal dispatch path that does the actual work.

The methodology we release is parameterised so that it generalises across these targets with minimal modification:

- The `cdm_arch_discover.py` discoverer locates a constructor’s vtable-installation pattern in any PE binary by structural search; the only Widevine-specific assumption is the export name `CreateCdmInstance`, which a researcher applying this to PlayReady or FairPlay substitutes with the relevant entry point.
- The shape-only shim’s hook semantics are independent of the per-slot signature; each `WvHookSlotN` takes the full callee-save register tuple. The same shape applies to a PlayReady or FairPlay vtable that uses a different signature shape per slot.
- The CFG bypass via `SetProcessValidCallTargets` is a Windows-process-level mitigation, not a Widevine-level one. It applies unchanged to any Windows process that loads a third-party library and enables CFG.

A separate paper on each of PlayReady and FairPlay applying this methodology is, we hope, follow-on work that future researchers will produce. The Buchanan model that constrains the methodology to shape-only observation is the same in all three cases.

Para investigadores que sigan

Two of the paper’s design choices are intended to short-circuit common dead ends in follow-on work.

First, the methodology is structured to be applied to any future `widevinecdm.dll` build with no manual RVA editing. The discoverer locates the outer vtable from the binary alone; the runtime instrumentation script takes the discovered RVA as a parameter. Re-running the same methodology against a future Chrome stable, or against the `libwidevinecdm.so` variants that ship on Linux and Chrome OS, is mostly a matter of running the discoverer.

Second, the false-start narrative in Section 5.7 is intended to head off two common dead ends that look superficially correct. Researchers entering this RE space should expect to encounter several internal nineteen-slot dispatch surfaces that are *not* the public CDM_11 vtable, and should expect the first-attempt obfuscation-attribution layer to be wrong by one indirection. The constructor-trace pivot from `CreateCdmInstance` is the reliable disambiguator.

14. Ética, disclosure, y lo que no hicimos

This study contains no:

- decrypted frame, in any form, captured to disk, sent over IPC, or retained transiently across the run of the shim;
- ciphertext byte, transient or persistent, retrieved through the shim;
- key material, content key, key wrap, or content-encryption key, captured at any layer;
- attack against `peauth.sys`, the Windows PMP, or any decoder-side component positioned to materialise a decoded frame;
- exploit code, weaponized shim, or off-the-shelf content-circumvention tool;
- production network attack against Netflix, Bitmovin, or any EME application studied;
- reverse-engineering result we cannot reproduce from the released source-only methodology against the same binary.

We have forensically verified the absence of content bytes in every artefact we produced. The captured shape logs are pure ASCII JSON totalling ~56 MB and contain only the metadata fields described in Section 6.3. The unique `key_id` hash count in the captured runtime events is zero: the code path that would have computed a hash of a Netflix `key_id` was never exercised at runtime, because the slots that take an `InputBuffer_2*` argument were not invoked.

No vendor was notified prior to this submission because the work does not constitute a vulnerability report. We will coordinate with Google’s Chrome Vulnerability Reward Program and the relevant Widevine security channels prior to any future submission of exploit-class findings derived from the methodology here.

15. Conclusión

Hemos presentado un modelo arquitectural verificado byte-a-byte del Widevine CDM_11 dispatch as it ships in two representative representatives de la familia Chromium, y un mapa empírico which CDM_11 slots are exercised during steady-state reproducción Netflix hardware-decoded en steady-state. Las ranuras por-cuadro de cara al usuario están dormidas bajo decode por hardware table generates $\sim 2.3k$ events per second of cryptographic or scheduler activity that is invisible from the EME boundary. The session-management slots remain reachable and exercise the same dispatch path regardless of whether the host EME application is Netflix or a non-commercial Widevine test source. The library distributes mixed-boolean-arithmetic obfuscation across the slot space in a pattern that mirrors the depth at which each slot begins to parse structured attacker-influenced data.

The work is situated in two decades of prior DRM research, follows the Buchanan model of architectural-only publication, and produces no extraction artefact. The methodology and all source needed to reproduce the empirical results are released alongside the paper.

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A. Tabla de despacho por ranura, ambos binarios

Slot	Method	Edge 150 RVA	Chrome 149 RVA
0	Initialize	0x1db390	0x1e0420
1	GetStatusForPolicy	0x1da7a0	0x1df8d0
2	SetServerCertificate	0x1dab10	0x1dfba0
3	CreateSessionAndGenerateRequest	0x1dad00	0x1dfdc0
4	LoadSession	0x1dad20	0x1dfde0
5	UpdateSession	0x1dad40	0x1dfe00
6	CloseSession	0x1dad60	0x1dfe20
7	RemoveSession	0x1dad80	0x1dfe40
8	TimerExpired	0x1dada0	0x1dfe60
9	Decrypt	0x1dad0	0x1dfe80
10	InitializeAudioDecoder	0x1dade0	0x1dfea0
11	InitializeVideoDecoder	0x1dae00	0x1dfec0
12	DeinitializeDecoder	0x1dae20	0x1dfee0
13	ResetDecoder	0x1dae40	0x1dff00
14	DecryptAndDecodeFrame	0x1dae60	0x1dff20
15	DecryptAndDecodeSamples	0x1dae80	0x1dff40
16	OnPlatformChallengeResponse	0x1daea0	0x1dff60
17	OnQueryOutputProtectionStatus	0x1daec0	0x1dff80
18	OnStorageId	0x1db310	0x1e03a0

B. Esquema del log de forma

The captured JSONL events use one of two shapes.

Generic call (for slots that are not InputBuffer_2-consuming):

```
1 {
2   "kind" : "call",
3   "slot" : <int 0..18>,
4   "n"    : <monotonic call number for this slot>,
5   "tsc"  : <QueryPerformanceCounter snapshot>,
6   "rh"   : <FNV-1a-32 hash of rcx XOR rdx XOR r8 XOR r9>
7 }
```

Shape (for slots 9, 14, 15):

```
1 {
2   "kind" : "ib",
3   "slot" : <9|14|15>,
4   "n"    : <monotonic call number>,
5   "tsc"  : <QueryPerformanceCounter snapshot>,
6   "es"   : <encryption_scheme; 0=clear, 1=CENC-CTR, 2=CBCS>,
7   "is"   : <data_size; size only, never the bytes>,
8   "kis"  : <key_id_size>,
9   "kh"   : <FNV-1a-32 hash of first 16 bytes of key_id>,
10  "ivs"   : <iv_size>,
11  "ivnz"  : <1 if IV has any non-zero byte, else 0>,
12  "ns"   : <num_subsamples>,
13  "pc"   : <pattern_crypt>,
14  "ps"   : <pattern_skip>,
15  "mt"   : <media timestamp>
16 }
```

Notably absent fields: data, key_id bytes, iv bytes, subsamples array, output-frame contents, content-encryption key, key-wrap key, OEMCrypto internal state. These are deliberately omitted.